

Refresh / Content Opportunity Scoring on Real Search Data: A Repeatable Baseline and Grouped Validation Study

Ahmed — Cairo, Egypt — FlyRank ML Internship — Lane: Refresh Opportunity Scoring — 2026-09-13
Built on FlyRank ML Internship dataset — <https://flyrank.ai/> — Warehouse Release v20260703

Abstract— Which pages should a content team review first for refresh when capacity is limited? We study FlyRank's pseudonymized warehouse release v20260703 (78.8M daily facts, 104 clients, 519k content items), iterating on mid-panel month March 2026 with April 2026 as future outcome window and June 2026 sealed as test. We verify two flag-linked signals: CTR vs position and volume vs clicks, both CONFIRMED with bucket tables and n. We encode a transparent baseline rule score = impressions/(position+1) with reason code STRIKING_DISTANCE_HIGH_IMPR and action label refresh_candidate, writing ranked queue to work/outputs/baseline_action_score.csv. We then train a Random Forest on 7 honest features with GroupKFold by client to avoid client-memorization. On honest grouped split (base rate 53.6% declining), baseline achieves AUC 0.627 and P@50 0.78, Logistic Regression 0.633/0.80, Random Forest 0.637/0.82 — model beats baseline by +1pp AUC and +4pp P@50 while passing leakage checks. We ship ranked refresh queue with reason codes and a public research paper with reproducibility receipts.

Keywords— Search Intelligence, Content Refresh, Baseline Model, GroupKFold, CTR, Learning to Rank, FlyRank

I. INTRODUCTION

In applied Search Intelligence, 'the right page to fix' means the right page to REVIEW FIRST, based on evidence and limited capacity — not a page guaranteed to recover if edited. Proving causality requires an experiment; this data alone provides decision-support. A good candidate has: enough demand to matter, evidence of movement/weakness, an action someone could take, reason codes a reviewer can inspect, no leakage, no obvious consolidation/seasonality/noise explanation. We confirm lane: Refresh / Content Opportunity Scoring.

II. DATA

Release flyrank_pseudonymized_warehouse_release_v20260703 — export 2026-07-03, daily facts through 2026-06-30. Tables: dim_clients 104 rows, dim_content 519,606 rows, fact_content_daily_performance 78,835,655 rows (grain report_date × client × content, partitioned by month), fact_content_query_90d 2.4M rows. Iteration month 2026-03, label window 2026-04 future, sealed test 2026-06. Columns: report_date, client_hash_id, content_hash_id, gsc_data_available, ga4_data_available, gsc_impressions, gsc_clicks, gsc_avg_position, ga4_sessions, month plus dim_content join for content_age_days, word_count via ANY_VALUE() never SUM. Excluded: Future April metrics (leakage), trend_direction/trend_pct (label-derived), product flags health_score/priority_score (circular), IDs as features (grouping only), query table 90d aggregates (window overlap), zero-filled GA4 rows where ga4_data_available IS NOT TRUE — must use IS TRUE (three-valued). Privacy: No client names, domains, URLs, private queries.

III. METHODOLOGY

Assumptions: Unbalanced panel — per-client history depth differs (9 of 70 clients have 12+ months). Prefer per-client windows for seasonality, but global March window used for v1 with filter gsc_data_available IS TRUE. Observational only — we rank candidates, not prove causal refresh impact. Volume floor impressions ≥ 100 to avoid noise. Features (7 honest): impressions_mar, clicks_mar, ctr_mar, avg_position_mar, sessions_mar, content_age_days, word_count — each knowable at decision moment March 31. Label: Proxy is_declining_next_month = 1 when April impressions < 0.8*March AND March ≥ 100, else 0. Base rate 53.6%. Baseline: Transparent rule score = gsc_impressions / (gsc_avg_position+1) filtered IS TRUE, impr ≥ 100, pos 5-20, reason_code STRIKING_DISTANCE_HIGH_IMPR, action_label refresh_candidate. Validation: GroupKFold by client_hash_id (5 folds) — tests on clients never seen. Time-aware: Feature March → Label April, strictly after decision moment. Leakage checks: No future April cols, no trend_pct, no health_score, IDs only grouping, query table excluded, flags filtered IS TRUE. Metrics: ROC AUC, Average Precision, Precision@K with base rate.

IV. RESULTS

On honest GroupKFold split (test 41k rows, base rate 53.6%): Baseline AUC 0.627 AP 0.646 P@50 0.78, LogReg 0.633/0.654/0.80, Random Forest 0.637/0.660/0.82 — model beats baseline by +1pp AUC and +4pp P@50. Top features: ctr_mar 28%, avg_position_mar 24.8%, content_age_days 22.5% — aligns with flag-linked signals checked. Leakage demo from W03: Adding future impressions_apr → AUC 0.642 → 0.987 (delta +0.345 = confession). Removed, honest kept.

V. LIMITATIONS

Unbalanced panel + three-valued flags: Per-client history varies, GA4 zero-filled with FALSE but millions NULL, must use IS TRUE. Feature frame biased to established tracking clients. Query table window overlap: fact_content_query_90d fixed 90-day window overlaps April label; only prev30 safe — excluded in v1. No causal proof: Observational only. Decline could be consolidation, seasonality, SERP feature change, or noise. We mitigate with volume floor ≥ 100 and persistence, but cannot prove refresh would cause recovery without experiment. Claims use careful words: observed, measured, directional, decision-support. Sample not random: _sample = June 2026 final month = natural outcome window, treated as sealed test.

VI. RANKED RECOMMENDATIONS

Top 10 from final ranked queue ($\text{final_score} = 0.7 * \text{model_prob} + 0.3 * \text{norm_baseline_score}$) with reason codes STALE_VISIBLE_PAGE, STRIKING_DISTANCE_HIGH_IMPR, etc. Full queue: work/outputs/refresh_queue.csv (100 rows sample). What would make a top pick wrong? Consolidation: sibling URL absorbed demand. Seasonality: topic peaks March. SERP change: impressions stable, clicks drop, CTR low $\rightarrow$ AI overview. Noise: days_with_impressions low (spike). Intent mismatch: low CTR due to informational query vs transactional page. Feedly type: missing keyword data pattern.

VII. REPRODUCIBILITY

Repo: <https://github.com/ahmed171102/Task-1>. Notebooks: w03_data_contract.ipynb, w04_baseline_score.ipynb, capstone.ipynb. Outputs: baseline_action_score.csv, refresh_queue.csv, capstone_metrics.json, roc_comparison.png. Run: pip install duckdb scikit-learn pandas matplotlib $\rightarrow$ set HF_TOKEN secret $\rightarrow$ Run all cells in capstone notebook. Paper source: docs/index.html (GitHub Pages) and capstone/paper.html — inline styles, embedded charts base64. IEEE version: docs/ieee.html and LaTeX capstone/paper_ieee.tex.

VIII. ACKNOWLEDGMENTS

Built on the FlyRank ML Internship dataset — pseudonymized warehouse release v20260703 (78.8M daily facts) hosted on Hugging Face at FlyRank/internship-warehouse, gated with instant approval. Data credit: FlyRank.ai — Applied Search Intelligence platform. <https://flyrank.ai/>

Method	ROC AUC	Avg Prec	P@50
Baseline (impr/(pos+1))	0.627	0.646	0.78
LogReg (7 feats)	0.633	0.654	0.80
RF (7 feats)	0.637	0.660	0.82

References: [1] FlyRank Warehouse Release v20260703, Hugging Face. [2] FlyRank.ai. [3] Pedregosa et al., Scikit-learn, JMLR 2011. [4] DuckDB.